hw14

1 Overview

The `Lahman` package contains data on pitching, hitting and fielding statistisc for MLB players from 1871 to present day. Our goal in this assignment is to explore the age at which players hit their "peak" performance. We'll specifically be exploring data on players who had long careers in the MLB (note that we won't be able to generalize our results to all players since we're focusing on those with long careers).

Note: Due to the more open-ended nature of this assignment, the autograder will not be used for this assignment.

Due date: Tuesday, Oct. 27

2 Details

You'll need to make use of the following data frames from this package:

- `Batting`: contains the batting info for each player and season
- `People`: contains info on each player

You'll want to take a look at the help pages for both data frames so you know what the column names are referring to.

Here's some code to get you started that takes care of some intial data processing:

```
library(Lahman)

# some players have multiple rows per season if they had stints with different
# teams. Combine these so we have one row per season.
bat <- aggregate(
  . ~ playerID + yearID,
  Batting[, !(names(Batting) %in% c("teamID", "stint", "lgID"))],
  sum
)
# SF (sacrifice flies) and HBP (hit by pitch) weren't recorded in older
# seasons - replace NA values with 0
bat$SF[is.na(bat$SF)] <- 0
bat$HBP[is.na(bat$HBP)] <- 0

ppl <- People
```

We need to use some metric to determine a players performance in a season. We'll use OPS (on-base plus slugging). You'll need to add the following columns to `bat` by calculating some stats from the existing columns.

- `SLG`: Slugging percentage; represents the total number of bases a player records per at-bat. This can calculated as follows: `(H - X2B - X3B - HR + 2 * X2B + 3 * X3B + 4 * HR) / AB`
- `OBP`: On-base percentage; represents how frequently a batter reaches base per plate appearance. Calculated as follows: `(H + BB + HBP) / (AB + BB + HBP + SF)`
- `OPS`: On-base plus slugging; adds on-base percentage and slugging percentage to get an overall measure of offensive production. Calculated as follows: `OBP + SLG`.

Note: the `with()` function can be used to save you some typing here. This function takes a data frame and an expression and evaluates the expression in the context of the data frame. This means that the name of the data frame doesn't have to be included. For example, consider the built-in `mtcars` data frame. I could create a new column as follows:

```
mtcars$newcol <- mtcars$mpg + mtcars$cyl + mtcars$hp + mtcars$drat
```

I could save some typing using `with()`:

```
mtcars$newcol <- with(mtcars, mpg + cyl + hp + drat)
```

You'll need to complete the following tasks:

1. Since we're trying to look at the age at which a player reaches their peak, we want to only include players with fairly long careers. Narrow down the players as follows:
    - players who are no longer playing (check the documentation of `People` to figure out how to determine this)
    - players who had at least 10 seasons with at least 200 at-bats (`AB`)
    - in the final batting data, only include seasons with at least 200 at-bats.
2. Notice that the `Batting` data frame doesn't include the players' age, so we'll need to create a column with their age. How can we do this? (*Hint: merging might be involved.*)
3. Create plots for at least two players that shows their OPS by age to get an idea of what these patterns can look like.
4. Write code to find the age at which each player had the "peak" OPS. Create a histogram showing these ages. Also calculate the mean value of these ages.

Note:

- Note that by filtering players to those with long careers, we're likely introducing bias into our data - players who "peaked" early may have left the league sooner, so we may be removing people with lower peak ages. These results should be interpreted with caution.
- This isn't necessarily the best way to analyze this data - there are more sophisticated ways of addressing this question.